

Comparison with Clique Number of Tournaments II

Detailed overlap between the June 6, 2026 preprint and the tournament clique-number repository

Repository checked through D38, June 7, 2026

Contents

Sources and scope	1
Common notation	2
Executive comparison	3
1. The $k=3$ result	4
2. The $k=4$ results	4
3. The new $k=5$ theorem in the repository	5
4. Conjecture 5.10 and Question 5.9	8
5. Proof-method comparison	9
6. Other repository results absent from the preprint	10
7. Major preprint results outside the repository's scope	10
8. Novelty and provenance cautions	11
9. Concise conclusion	11

Last checked against the repository on June 7, 2026 (through D38).

Sources and scope

This note compares:

- Guillaume Aubian and Samuel Coulomb, *Clique Number of Tournaments II*, preprint dated June 6, 2026 ([local PDF](#));
- the results recorded in this repository under `problems/tournament_clique_number_omega_cluster/`, in particular:
 - the $k=3$ proof;
 - the value proof for `AC_n[C3]`;
 - the deletion proof for `AC_n[C3]`;
 - the $k=5$ proof for `AC_n[AC_n]`;
 - the unified $k=3, 4, 5$ account;
 - the refutation of the proposed substitution upper bound;
 - the H16 certificate generator and `certificate manifest`;
 - `ledger.json`, which is the authoritative status record.

The preprint's critical-tournament results are in Section 6 (printed pages 16-17). Its complexity theorem is Theorem 1.1/Section 3, its approximation results are in Section 5, and its `chi_vec`-bounding results are in Section 4.

The authoritative ledger now extends through D38. P20 remains promoted to `can_enter_proved=true`, so Conjecture 5.10 is proved in the repository for $k=3, 4, 5$. The mathematical frontier has not moved beyond D35: no new value of k has been settled. Decisions D36-D38 instead strengthen the record by completing omitted $k=5$ casework, exporting independently checkable certificates for the H16 counterexample, correcting the certificate encoding, and systematically removing claims that exceeded the scope of sampled data.

The main comparison concerns directed clique number, critical tournaments, Conjecture 5.10, and Question 5.9. The preprint is substantially broader, so its complexity, approximation, and `chi`-boundedness results are summarized separately near the end.

Common notation

For a tournament T and a total order prec of its vertices, the backedge graph T^{prec} has an edge uv , with $u \prec v$, when the tournament arc is $v \rightarrow u$. The directed clique number is

$$\omega_{\text{vec}}(T) = \min_{\text{prec}} \omega(T^{\text{prec}}).$$

A tournament is k - ω_{vec} -critical when

$$\omega_{\text{vec}}(T) = k$$

and

$$\omega_{\text{vec}}(T-v) = k-1 \text{ for every vertex } v.$$

For odd $N=2m+1 \geq 7$, the repository writes

$$\text{AC}_N = \text{Cay}(\mathbb{Z}/N, \{1, \dots, m-1\} \cup \{m+1\}).$$

The preprint's tournament $T_{\{2n+1\}}$ in Proposition 6.1 has generator set

$$\{1, \dots, n-1\} \cup \{n+1\}.$$

Thus, after setting $N=2n+1$, the two definitions are identical:

$$T_{\{2n+1\}} = \text{AC}_{\{2n+1\}}.$$

For lexicographic substitution, $S[H]$ means that every vertex of the outer tournament S is replaced by a copy of the inner tournament H . Therefore

$$\Delta(H, H, H) = C_3[H].$$

This direction matters. In general,

$$C_3[\text{AC}_N] \neq \text{AC}_N[C_3] \neq \text{AC}_N[\text{AC}_N].$$

The first two have the same order $3N$, but the definitions use opposite substitution directions. A direct isomorphism check made for this comparison gives non-isomorphism for $N=7, 9, 11$; no all- N non-isomorphism theorem is needed below.

Executive comparison

Level	Preprint	Repository	Relationship
k=3	Proposition 6.1: AC_N is 3-critical for every odd $N \geq 7$	P13: the same family and theorem	Exact overlap in statement and construction; different proofs
k=4	Proposition 6.2: $C3[AC_N] = \Delta(AC_N, AC_N, AC_N)$ is 4-critical	P16+P18: $AC_N[C3]$ is 4-critical	Same conclusion, opposite product direction; checked non-isomorphic for $N=7, 9, 11$
Earlier repo k=4 route	The preprint proves the Delta family	G22 had found and verified exactly this Delta family but lacked a valid uniform proof	The preprint supplies the missing theorem for the old G22 candidate
k=5	No explicit or unconditional 5-critical family	P19+P20: $AC_N[AC_N]$ is 5-critical for every odd $N \geq 7$; omitted casework repaired in D36-D38	New result relative to the supplied preprint
General substitution	No additive formula for $\omega_vec(S[H])$ is claimed	D35 refutes $\omega_vec(S[H]) \leq \omega_vec(S) + \omega_vec(H) - 1$; D36-D37 add proof-producing CNF/DRAT artifacts	New structural result in the repository; no contradiction with the preprint
k>=6	Conditional infinitude unless $P=NP$; no explicit family	Open; no explicit 6-critical witness is recorded, H16 is closed, and direct searches give negative evidence on explicitly limited scopes	Neither source gives an unconditional explicit family
Question 5.9	Negative at k=3, 4 follows from Propositions 6.1 and 6.2, although not formulated that way	Explicitly proved negative at k=3, 4, 5	Repository adds the k=5 negative result and spells out the monotonicity argument

1. The $k=3$ result

Preprint result

Proposition 6.1 states that, for every integer $n \geq 3$,

$$T_{\{2n+1\}} = AC_{\{2n+1\}}$$

is 3- ω_{vec} -critical.

Its proof has two main inputs.

1. For an arbitrary vertex order, take its last vertex v_i . Several specified out-neighbours of v_i are adjacent to it in the backedge graph. Three of those neighbours form a directed triangle, so two are adjacent in every backedge graph. Together with v_i , they give a backedge triangle. This proves $\omega_{\text{vec}}(AC_N) \geq 3$.
2. The proof invokes the theorem of Neumann-Lara and Urrutia that this same tournament is 3-dichromatic-critical. Since $\omega_{\text{vec}} \leq \text{dichromatic number}$, every vertex deletion has directed clique number at most 2. This supplies the deletion upper bound used for criticality.

The proof is short and conceptually driven by the known dichromatic-critical family.

Repository result

P13 proves the same theorem:

$$AC_N \text{ is } 3\text{-}\omega_{\text{vec}}\text{-critical for every odd } N \geq 7.$$

The repository proof is more explicit about the directed clique number itself.

1. In the identity order, every backedge gap is at least m , so a backedge clique has size at most 3.
2. The only backedge triangle in this order is $\{0, m, 2m\}$.
3. The lower bound is obtained from $\text{dom}(AC_N) \geq 3$ and the known inequality $\text{dom}(T) \leq \omega_{\text{vec}}(T)$. The domination statement is reduced to and proved by the autocorrelation identity

$$\min_{\{t \neq 0\}} |N_0 \cap (N_0 + t)| = 2.$$

4. Deleting 0 removes the unique identity-order backedge triangle, giving $\omega_{\text{vec}}(AC_N - 0) \leq 2$; an explicit surviving directed triangle gives the reverse inequality.
5. Vertex-transitivity extends the deletion statement to every vertex.

Exact overlap and differences

The theorem and the family are exactly the same. The proof dependencies differ:

- the preprint relies on the previously known 3-dichromatic-criticality of AC_N ;
- the repository gives a direct backedge-order proof of the upper and deletion bounds, plus a domination/autocorrelation proof of the lower bound.

The repository also explicitly derives the failure of Question 5.9 at $k=3$. Every proper subtournament of a 3-critical AC_N lies in some deletion $AC_N - v$, hence has directed clique number at most 2. Since $|AC_N|$ is unbounded, no function $e_{\ell}(3)$ can exist.

This failure is also an immediate consequence of the preprint's Proposition 6.1, but the preprint does not formulate the result in the language of Question 5.9.

2. The $k=4$ results

Preprint construction: $C3[AC_N]$

Proposition 6.2 proves that

$$\Delta(\mathcal{AC}_N, \mathcal{AC}_N, \mathcal{AC}_N) = C3[AC_N]$$

is 4- ω_{vec} -critical for every odd $N \geq 7$.

For the lower bound, take the first vertex of an arbitrary ordering, say in block A. Every vertex of the preceding cyclic block C is adjacent to it in the backedge graph. Since $\omega_{\text{vec}}(C) = 3$, this forces a backedge K_4 .

For deletion, suppose x is deleted from block A. Choose vertices b and c in the other two blocks, and use 2-dicolourings

$(A1, A2), (B1, B2), (C1, C2)$

of $A-x$, $B-b$, and $C-c$. The preprint orders the remaining vertices as

$b < c < A1 < B1 < C1 < A2 < B2 < C2$

with topological orders inside the colour classes. It then checks that the resulting backedge graph is K_4 -free. Thus every deletion has directed clique number 3.

The construction uses the fact that AC_N is 3-dichromatic-critical, not only 3- ω_{vec} -critical.

Repository construction: $AC_N[C3]$

P16 and P18 prove that the reverse product

$AC_N[C3]$

is 4- ω_{vec} -critical for every odd $N \geq 7$.

The value proof uses:

- the general lexicographic lower bound

$\omega_{\text{vec}}(S[H]) \geq \omega_{\text{vec}}(S) + \omega_{\text{vec}}(H) - 1;$

- a merged order with key $c(t) + d(h);$
- a key-class analysis excluding a backedge K_5 .

The deletion proof uses a different order, $d_{\text{then}}c$, with five bands. Its case analysis establishes:

number from the h in $\{1, 2\}$ layers
 + number from the $h=0$ layer
 ≤ 3 .

The difficult case is the $(2, 2)$ configuration, ruled out by the incompatibility

$\text{not}(1 + \delta \text{ in } g \text{ and } m + 1 + \delta \text{ in } g).$

Relationship to the old G22 candidate

Before the final $AC_N[C3]$ proof was obtained, the repository investigated

$D_N = \Delta(AC_N, AC_N, AC_N).$

This is exactly the preprint's $C3[AC_N]$ family. The repository verified $\omega_{\text{vec}}(D_N) = 4$ and the deletion value 3 over a substantial finite range, but recorded the proposed uniform proof as G22 because:

- the claimed Δ upper-bound lemma did not exist in the cited source;
- the uniform deletion argument had not been supplied;
- finite verification was correctly not treated as a theorem.

Proposition 6.2 of the preprint supplies the missing rigorous proof. Thus the preprint does not merely reach the same $k=4$ conclusion: it validates an exact construction that appears in the repository's historical research record.

What is genuinely different

The final repository theorem and the preprint theorem are not duplicate constructions:

preprint: $C3[AC_N] =$ three large AC_N blocks in a directed 3-cycle;
 repository: $AC_N[C3] = N$ small $C3$ blocks arranged according to AC_N .

They have the same order $3N$ and are both vertex-transitive and 4-critical, but they use opposite substitution directions and different proof mechanisms.

Consequently there are two explicit infinite 4-critical families available from the combined record.

As at $k=3$, either family directly implies that Question 5.9 fails at $k=4$. The repository states and proves this consequence explicitly.

3. The new $k=5$ theorem in the repository

Statement

P19+P20 prove:

For every odd $N=2m+1 \geq 7$, the lexicographic product $AC_N[AC_N]$ is 5- ω_{vec} -critical.

Equivalently,

$$\omega_{\text{vec}}(AC_N[AC_N]) = 5$$

and, for every vertex v ,

$$\omega_{\text{vec}}(AC_N[AC_N]-v) = 4.$$

The orders N^2 are unbounded, so this is an infinite explicit family. It proves Conjecture 5.10 at $k=5$ and gives a negative answer to Question 5.9 at $k=5$.

Role of P19

P19 is the finite seed and discovery certificate:

$$AC_7[AC_7]$$

has order 49, directed clique number 5, and every vertex deletion has directed clique number 4.

The computation isolated two useful orders:

- the merged-sum order for the full tournament;
- the inner_then_outer order for a vertex deletion.

P19 alone is an explicit 5-critical tournament and gives the finite lower bound $\ell(5) \geq 49$, but a single tournament does not prove Conjecture 5.10.

Role of P20

P20 turns the P19 pattern into a uniform theorem for all odd $N \geq 7$.

Let a vertex of $AC_N[AC_N]$ be (a, b) , with outer coordinate a and inner coordinate b . Use the three-level potential

$$c(t) = \begin{cases} 3 & \text{if } t=0, \\ 2 & \text{if } 1 \leq t \leq m, \\ 1 & \text{if } m+1 \leq t \leq 2m. \end{cases}$$

The main proof components are as follows.

Full-tournament lower bound The general lexicographic lower bound gives

$$\begin{aligned} \omega_{\text{vec}}(AC_N[AC_N]) &\geq \omega_{\text{vec}}(AC_N) + \omega_{\text{vec}}(AC_N) - 1 \\ &= 3 + 3 - 1 \\ &= 5. \end{aligned}$$

Deletion lower bound After deleting $(0, 0)$, the remaining tournament contains

$$(AC_N - 0)[AC_N].$$

Therefore

$$\begin{aligned} \omega_{\text{vec}}(AC_N[AC_N] - (0, 0)) &\geq \omega_{\text{vec}}(AC_N - 0) + \omega_{\text{vec}}(AC_N) - 1 \\ &= 2 + 3 - 1 \\ &= 4. \end{aligned}$$

Deletion upper bound Order the surviving vertices by

$$(c(b), c(a), a, b).$$

The pair $(c(b), c(a))$ defines one of eight surviving cells in

$$\{1, 2, 3\}^2 \text{ minus } \{(3, 3)\}.$$

The proof first shows that a backedge clique contains at most one vertex from each cell. It then rules out every five-cell clique. The obstruction is organized as 20 listed infeasible cell sets:

- 10 triples;
- 4 outer-source quadruples;

- 6 square quadruples.

They are closed by two symbolic mechanisms.

1. **Forced-value chains.** The almost-consecutive generator arithmetic forces outer coordinates to $m+1$, to m , or into an interval disjoint from g , after which another required backward arc is impossible.
2. **Square obstruction and Lemma H17.** For x in $H=[m+1, 2m]$ and y in $L=[1, m]$, the common neighbourhood

$$N^-(x) \cap N^-(y)$$

lies in a single band. More precisely, it lies in $[0, m-1]$ when $x-y \leq m$, and in $[m+1, 2m-1]$ when $x-y \geq m+1$. It therefore cannot contain simultaneously the high-band and low-band representatives required by a square configuration.

In the repaired proof, the ten triples and four outer-source quadruples each have an explicit residue chain. The six square quadruples are handled by their common alternating high/low-band pattern: a same-block equality branch contradicts tournament antisymmetry, while the all-distinct branch contradicts Lemma H17. The argument needs exactly these two facts:

1. each of the 20 listed cell sets is symbolically infeasible;
2. every one of the $C(8, 5)=56$ five-cell subsets contains at least one listed obstruction.

It does not require the 20 sets to be the exact or inclusion-minimal obstructions.

This proves

$$\omega_{\text{vec}}(\text{AC_N}[\text{AC_N}] - (\emptyset, \emptyset)) \leq 4.$$

Together with the lower bound, the deletion value is exactly 4. Vertex-transitivity gives the same value for every deletion.

Full-tournament upper bound The value upper bound is a corollary of the deletion argument. In the chosen cell order, $(\emptyset, \emptyset)$ is the unique $(3, 3)$ vertex. A backedge clique either:

- omits $(\emptyset, \emptyset)$, and hence has size at most 4; or
- contains $(\emptyset, \emptyset)$, and has at most four additional vertices.

Thus the full backedge graph has clique number at most 5, completing

$$\omega_{\text{vec}}(\text{AC_N}[\text{AC_N}]) = 5.$$

Verification and red-team status

The proof is not being promoted from finite data alone.

- The finite computations discovered the cell structure and checked the obstruction list.
- The final proof supplies uniform residue arguments and the interval proof of Lemma H17.
- Seven adversarial checks reproduced the load-bearing claims.
- The red-team found one false band-L arc statement. The correct statement is

$$-a \text{ in } g \text{ iff } a=m, \text{ for } a \text{ in } [1, m].$$

The affected mirror arguments were rewritten using the forced value $a=m$.

- After correction, P20 was promoted to `can_enter_proved=true` in D28.
- D36 then found a genuine incompleteness in the written proof: the four outer-source quadruples had been asserted by “the same chains” with finite checks standing in for a uniform derivation, and the description of the six squares was inaccurate. The proof file now writes all four chains explicitly and gives the correct alternating-band argument for all six squares.
- D37-D38 removed the unsupported claim that the 20-set list was the exact minimal obstruction list. The theorem only uses symbolic infeasibility of the listed sets and the finite, n -independent coverage of the 56 five-subsets. A mechanical scope-overclaim lint now reports zero flags across the ledger and Markdown documentation.

Thus the current post-D36 proof is complete. The later decisions are proof and scope repairs, not a new construction or a change in the theorem’s statement.

Comparison with the preprint

The supplied preprint has no explicit 5- ω_{vec} -critical tournament and no unconditional infinite 5-critical family.

Its final observation says that the $k=4$ Delta construction generalizes if one starts from a k -dichromatic-critical tournament T satisfying $\omega_{\text{vec}}(T)=k$. The authors explicitly state that they do not know whether such tournaments exist for $k \geq 4$.

The repository's $\text{AC}_N[\text{AC}_N]$ theorem bypasses that missing input:

- it does not require a 4-dichromatic-critical base;
- it uses two copies of the 3-critical almost-consecutive family;
- its criticality proof is based on a product-specific cell order and residue structure.

Therefore P19+P20 are genuinely beyond the explicit critical-tournament results in the supplied preprint.

4. Conjecture 5.10 and Question 5.9

Unconditional explicit results

Combining the two sources gives:

k	Explicit infinite family	Source
3	AC_N	Both preprint Proposition 6.1 and repository P13
4	$\text{C3}[\text{AC}_N]$	Preprint Proposition 6.2
4	$\text{AC}_N[\text{C3}]$	Repository P16+P18
5	$\text{AC}_N[\text{AC}_N]$	Repository P19+P20

Thus Conjecture 5.10 is unconditionally proved for $k=3, 4, 5$.

For each displayed family, criticality gives a direct negative answer to Question 5.9 at that value of k : every proper subtournament lies in a one-vertex deletion and therefore has directed clique number at most $k-1$.

The repository explicitly records:

no $\text{ell}(3)$, no $\text{ell}(4)$, and no $\text{ell}(5)$.

Conditional result from complexity

Theorem 1.1 of the preprint proves that, for every fixed $r \geq 3$, deciding

$\omega_{\text{vec}}(T) \leq r$

is NP-complete.

Section 6 uses this to state that, for every $k \geq 4$, there must be infinitely many k - ω_{vec} -critical tournaments unless $\text{P}=\text{NP}$. The reasoning is that if only finitely many critical obstructions existed, testing for their induced presence would give a polynomial recognition algorithm for the corresponding threshold.

This is important evidence for the full conjecture, but it differs from the repository theorems in three ways:

- it is conditional on $\text{P} \neq \text{NP}$;
- it is nonconstructive;
- it does not identify a family or prove criticality of explicit tournaments.

The repository results for $k=3, 4, 5$ are unconditional and constructive.

Remaining gap

Neither source proves Conjecture 5.10 unconditionally for every $k \geq 3$.

The repository's current open range is $k \geq 6$, and no explicit 6- ω_{vec} -critical tournament is currently recorded. In the repository's internal terminology, the $k=6$ witness slot is still empty; this is not a claim that Question 5.9 has a positive answer at $k=6$. The general substitution lower bound

$\omega_{\text{vec}}(S[H]) \geq \omega_{\text{vec}}(S) + \omega_{\text{vec}}(H) - 1$

is proved, but the formerly proposed matching upper bound is **false**. An order-7 tournament H satisfies


```
omega_vec(C3) = omega_vec(H) = 2,
omega_vec(C3[H]) = 4,
omega_vec(H[C3]) = 3.
```

The product $C3[H]$ is itself 4-critical: every one-vertex deletion has directed clique number exactly 3. It is also minimal for fixed outer factor $C3$: an exhaustive isomorphism-class census finds no counterexample with $|H| \leq 6$ and exactly three among the 456 order-7 classes.

D36-D37 upgraded the computational lower bound to an independently checkable certificate package. Two distinct no-K4 ordering encodings, consecutive-chain and all-pairs-backward, each produce a DIMACS CNF with 210 variables, 10194 clauses, and 2214 forbidden transitive four-sets. For each encoding the repository stores:

- the DIMACS CNF and its SHA-256 checksum;
- a Cadical DRAT refutation;
- a redundant UNSAT result from both Cadical153 and Minisat22.

D37 caught that the first exported formulas used the reversed order direction. The literal direction was corrected and all CNFs, checksums, DRAT proofs, and solver results were regenerated. The current artifacts are in [data/h16_cert/](#).

Thus substitution is directional and $\omega_{vec}-1$ is not additive. The valid general upper bound is only

```
omega_vec(S[H]) <= omega_vec(S) * omega_vec(H),
```

obtained from block-respecting optimal orders; the counterexample attains equality $2*2=4$. The earlier block-laminarity lemma remains true, but it does not prove additivity: the induced block and within-block orders are arbitrary, whereas ω_{vec} is a minimum over all orders. The simple merged potentials for $AC_N[C3]$ and $AC_N[AC_N]$ remain valid special arithmetic constructions, but they cannot be promoted to a universal law.

The preprint's proposed generalization encounters a different-looking but related shortage: it would need suitable k -dichromatic-critical tournaments with directed clique number exactly k for $k \geq 4$.

New $k=6$ search information after D28

The newer repository rounds D29-D38 do not prove the $k=6$ case. D29-D35 provide the search results below; D36-D38 correct their quantifier and scope wording:

- no explicit 6-critical tournament or family has been found;
- in a sample of 42 single-orbit circulants of orders 37 through 49 whose identity order has clique number 6, every sampled tournament admits an order of clique number at most 5; this was not an exhaustive census of those orders;
- for the domination-number route, the exhaustive order-37 circulant census has maximum domination number 4; additional samples through order 65 also top out at 4, and the checked Paley tournaments through order 251 have domination number at most 5. Only the order-37 statement is exhaustive, so these computations do not prove that domination number 6 is absent below order 67;
- QR_67 remains unresolved with $\omega_{vec}(QR_{67})$ in $\{5, 6\}$;
- the exact positive value $\omega_{vec}(QR_{19}[AC_7])=6$ is known, but 6-criticality has not been established, so it does not provide the missing witness;
- the iterated outer- $C3$ Delta tower fails criticality at its first relevant test: all deletions of the order-27 level retain value at least 4;
- the old side claim that domination number at least 4 forces a Paley circulant is false; many off-Paley examples occur, but their domination number still stops at 4, so this correction does not open a $k=6$ route.

Accordingly, the live repository directions are now a special arithmetic criticality-lifting theorem tailored to the AC_N constructions, or a genuinely new parametric family with direct upper, lower, and deletion certificates. None of these negative computations changes the preprint comparison: its conditional complexity argument remains valid, but it gives neither an explicit family nor a construction at $k=6$.

5. Proof-method comparison

Issue	Preprint method	Repository method
$k=3$ lower bound	Last vertex plus a directed triangle among selected neighbours	Domination number plus an exact autocorrelation lemma

Issue	Preprint method	Repository method
k=3 deletion upper bound	Imported 3-dichromatic-criticality	Unique identity-order backedge triangle
k=4 construction	Outer C3, inner AC_N	Outer AC_N, inner C3
k=4 deletion	Six acyclic colour classes arranged in a global order	Five explicit potential bands and residue casework
k=5	No construction	Outer and inner AC_N; eight-cell deletion analysis
General product tool	Generalization from suitable dichromatic-critical inputs	General lex lower bound and product upper bound; additive upper bound refuted; sharper upper bounds only for special arithmetic products
Computational role	Small Python checks appear in gadget sections	Exact oracle, SAT encodings, template search, symbolic proof and red-team, proof-producing CNF/DRAT artifacts, and scope-overclaim lint

The two approaches are complementary. The preprint exploits dichromatic criticality to obtain concise proofs. The repository develops direct ω_{vec} orderings and arithmetic certificates, which makes the $k=5$ construction possible without a higher dichromatic-critical base.

6. Other repository results absent from the preprint

Within this problem folder, the repository also contains:

- a general proof of the lexicographic lower bound

$$\omega_{\text{vec}}(S[H]) \geq \omega_{\text{vec}}(S) + \omega_{\text{vec}}(H) - 1;$$
- explicit isolated 4-critical circulants, including Paley(19) and an order-21 circulant;
- a verified 5-critical cyclic product $AC_7[AC_9]$;
- exact branch-and-bound and SAT encodings for directed clique thresholds;
- extensive negative data on naive Paley, repeated-C3, and nested-product generalizations;
- the explicit order-7 refutation of H16, including the quantifier error in the earlier block-laminarity reduction;
- two independently encoded no-K4 CNFs with checksums and DRAT refutations for the H16 counterexample;
- a new 4-critical tournament $C3[H]$ of order 21, whose two width-2 factors attain the universal multiplicative upper bound.

These results are not claimed in the supplied preprint. Some are computational constructions or research diagnostics rather than general theorems; the ledger marks that distinction explicitly.

7. Major preprint results outside the repository's scope

The preprint proves or develops several results not addressed by this folder.

Complexity

- Theorem 1.1: for every fixed $k \geq 3$, deciding $\omega_{\text{vec}}(T) \leq k$ is NP-complete.
- Theorem 3.8 gives the base $k=3$ reduction from 3-SAT.
- Computing directed clique number is therefore NP-hard.
- The fixed threshold $k=2$ remains open in the preprint.

chi-bounding tournaments

The preprint studies the tournament analogue of the Gyrfás-Sumner program. Among its contributions:

- U_5 is shown to be χ_{vec} -bounding;
- a family of ordered graphs M_n is proved χ -bounding;
- necessary conditions for a tournament to be χ_{vec} -bounding are derived;
- explicit counterexamples are given to the conjecture that admitting a forest backedge graph is sufficient for being χ_{vec} -bounding;
- additional counterexamples are derived from incomparable ordered graphs and the Blanche Descartes construction.
- using an announced theorem that ordered matchings are χ -bounding, the preprint deduces that every class of tournaments of bounded twin-width is χ_{vec} -bounded.

None of this is a target of the present tournament-criticality folder.

Approximation and bounded certificates

Using the Crew-Fan-Koerts-Moore-Spirkl unavoidable-family theorem, the preprint records:

- a non-identity local-to-global certificate theorem with functions $f(k)$ and $ell(k)$;
- a polynomial-time gap algorithm certifying either $\omega_{\text{vec}}(T) \geq k$ or $\omega_{\text{vec}}(T) < f(k)$;
- an explicit algorithmic result that, when $\omega_{\text{vec}}(T) \leq 2$, constructs an ordering whose backedge graph has clique number at most 100.

These statements are compatible with the repository's negative answers to Question 5.9. The preprint's certificate theorem permits a threshold $f(k)$ strictly larger than k ; the critical families prove that the identity threshold cannot work for $k=3,4,5$.

Perfect tournaments

The preprint observes that the family AC_N is perfect in the directed sense used there and raises the classification of perfect tournaments as a question. This is not developed in the repository.

8. Novelty and provenance cautions

This comparison is mathematical, not a priority determination.

- The supplied PDF is dated June 6, 2026.
- The repository proof artifacts were edited on June 6-7, 2026.
- The preprint credits the 3- and 4-critical families to Aboulker, Charbit, and Thomassé.
- The repository contains an automated/human research history, including candidates found computationally before they were promoted to theorems.

The safe claims relative to the supplied materials are:

1. The repository's P13 and preprint Proposition 6.1 prove the same $k=3$ family.
2. Preprint Proposition 6.2 proves the exact Delta family recorded earlier as repository candidate G22.
3. The repository's final $k=4$ family $AC_N[C3]$ is different from the preprint's $C3[AC_N]$.
4. The explicit infinite $k=5$ family $AC_N[AC_N]$ does not appear in the supplied preprint.
5. The D35 counterexample to additive substitution, and the resulting order-21 4-critical tournament, do not appear in the supplied preprint; the preprint itself does not assert the refuted law.

Any stronger statement such as "first in the literature" requires a separate literature and chronology check.

9. Concise conclusion

The overlap is exact at $k=3$, and the preprint also proves one of the repository's earlier $k=4$ candidate families. The repository contains a second, reverse-product $k=4$ family. Its main additional theorem is the now-complete, post-D36-repaired P19+P20 result:

$AC_N[AC_N]$ is 5- ω_{vec} -critical for every odd $N \geq 7$.

Hence the combined unconditional picture is:

Conjecture 5.10 is proved for $k=3,4,5$,
 Question 5.9 fails at $k=3,4,5$,
 the universal additive substitution route is false,
 no explicit 6-critical witness is currently known,
 and the explicit/unconditional problem remains open for $k \geq 6$.